AJITHAR A

AI/ML SOFTWARE ENGINEER

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PROFILE SUMMARY

AI Software Engineer with 2.5 + years of experience developing AI powered backend services and integrating machine learning workflows into production applications. Hands on experience with Python, FastAPI, computer vision, OCR, LLM integration, speech processing, asynchronous APIs, and real time WebSocket systems. Experienced in leading AI development for assigned projects, mentoring junior developers, and delivering AI/ML training. Focused on building reliable, maintainable AI solutions that address practical business requirements.

EXPERIENCE

AI/ML Software Engineer — **DoubleTap Innovations and Technologies Pvt. Ltd. | Chennai,**
Jun 2024 – Present

- Develop and maintain AI powered backend features across recruitment, retail automation, wellness, video preparation, and task management applications.
- Lead the AI workstream for an assigned project, coordinate implementation activities, and guide junior developers on technical tasks and development practices.
- Integrate machine learning models and LLM APIs into FastAPI services, including input validation, structured responses, asynchronous workflows, and error handling.
- Build and troubleshoot real time processing workflows using WebSockets, including image frame processing, session management, and event handling.
- Develop AI token usage and cost reporting workflows covering usage aggregation, credit monitoring, cost analysis, and Excel report generation.
- Investigate production issues, validate end to end integrations, and support deployment and release workflows using Docker.

AI Software Intern — **TabSquare InfoTech Private Limited | Chennai,**
April 2024 – May 2024

- Assisted with software development, application testing, and debugging to support reliable application functionality.
- Worked with existing application workflows and backend components to understand feature implementation and system behaviour.
- Supported API testing and validation to identify issues and verify expected responses.
- Collaborated with team members to troubleshoot technical issues and complete assigned development tasks.

Technician — **Lampex Electronics Limited | Chennai, Jun 2019 – Jun 2021**

- Operated manufacturing equipment involved in the production and assembly of LCD/LCM display modules and electronic components.

Junior Technician — **Tenneco Automotive India| Hosur, Jun 2018 – Jun 2019**

- Operated manufacturing machines used in the production of automotive shock absorbers and suspension components for vehicle manufacturers.

PROJECTS

Project Name : Fulupo — Retail Automation Live Website | App Link

Tech Stack: Python, FastAPI, YOLOv8, OpenCLIP, OCR, LLM APIs, WebSockets, MongoDB, Docker.

- Built computer vision pipelines for product detection, tracking, image quality, cropping, best-frame selection, and catalog matching using visual attributes, embeddings, and OCR.
- Developed receipt/bill extraction, column detection, tax and product parsing, structured data processing, and Excel report generation.
- Implemented voice transcription, regional language search, inventory matching, image based product recognition, personalized recommendations, and AI recipe generation.
- Built real time grocery recognition and 7 segment weighing scale OCR using WebSockets and REST APIs, with session management and weight normalization; contributed to bank passbook and cheque OCR.

Project Name : InternSharks.ai — AI Recruitment Platform Live Website | App Link

Tech Stack : Python, FastAPI, LLM APIs, OCR, WebSockets.

- Developed resume processing workflows for PDF, DOCX, and image inputs, integrating text extraction, OCR, AI powered resume enhancement, ATS scoring, job description matching, and structured JSON analysis.
- Built an AI powered resume creation chatbot and contributed to candidate profiling, student prediction, personalized learning path generation, MCQ based skill assessments, and automated answer validation.
- Contributed to six rounds of AI powered mock interview workflows and single agent and multi agent group discussions, supporting candidate evaluation, interactive interview sessions, and assessment driven feedback.

Project Name : Selfietoons — AI Powered Video Generation Backend Live Website | App Link

Tech Stack : Python, FastAPI, OpenAI APIs, Whisper, MongoDB, Cloudflare R2.

- Developed AI workflows for character image generation, audio transcription, transcript interpretation, concept-image analysis, and AI generated source images for video preparation.
- Implemented multilingual speech to text processing using Whisper, transcript normalization, dialogue preparation, and model specific video prompt generation for Grok and Kling based on lip sync requirements.
- Built asynchronous processing workflows with request tracking, status polling, timeout handling, MongoDB based processing status, and Cloudflare R2 integration for generated image storage.

Project Name : Toice — AI Task Reporting Live Website | App Link

Tech Stack : Python, FastAPI, OpenAI APIs, MongoDB, Speech-to-Text, WebSockets.

- Developed audio transcription and AI powered task creation, extracting task details, assignees, priorities, and deadlines from voice inputs.
- Implemented AI driven task report generation by summarizing task records, chat messages, attachments, and activity history.
- Integrated LLM based information extraction and structured task processing to convert unstructured inputs into actionable tasks and organized reports.

Project Name : WellBits — AI Backend &Wellness Analysis Live Website | App Link

Tech Stack : Python, FastAPI, OpenAI API, OpenCV, NumPy, MongoDB, Redis, Cloudflare R2, WebSockets.

- Developed AI powered wellness APIs for meal image analysis, personalized nutrition and goal recommendations, food suggestions, meal scheduling, wellness challenges, and conversational AI wellness assistance.
- Worked on the ChewTracker pipeline for bite and chewing detection, including multiface session handling, real time WebSocket events, pause/resume workflows, session state management, and prediction validation against ground-truth data.
- Integrated MongoDB and Cloudflare R2 for data and media storage, AI token and cost tracking, request validation and concurrency controls, and Docker based deployment.

Project Name : AI Token and Cost Management System Live Website

Tech Stack : Python, FastAPI, AI API's, MongoDB, Docker,

- Developed token-usage aggregation, AI cost reporting in USD and INR, credit monitoring, and low credit email alerts.
- Implemented natural language analytics, Excel report generation, and role based access controls.

LEADERSHIP, TRAINING & INDUSTRY ENGAGEMENT

- **AI Project Lead**: Led AI development activities for an assigned project and mentored junior developers.
- **Hackathon Judge** — SRM College, Chennai: Evaluated participant solutions at an AI related hackathon.
- **AI/ML Trainer** — DoubleTap Innovations and Technologies: Delivered AI/ML training to interns and course participants.

EDUCATION

Bachelor of Technology - Artificial Intelligence and Data Science
Mepco Schlenk Engineering College
2021 – 2024

Diploma - Electrical and Electronics Engineering
Government Polytechnic College
2015 – 2018

CERTIFICATIONS

- Natural Language Processing – NPTEL (IIT Kharagpur)
- Data Science for Engineers – NPTEL (IIT Madras)
- Introduction to Machine Learning – NPTEL (IIT Kharagpur)